

1.

```
main.c
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char str[100];
7     int i, n, valid = 1;
8     printf("Enter the string: ");
9     scanf("%s", str);
10    n = strlen(str);
11    for (i = 0; i < n; i++)
12    {
13        if (str[i] != '0' && str[i] != '1')
14        {
15            valid = 0;
16            break;
17        }
18    }
19    if (valid && n >= 2 && str[0] == '0' && str[n - 1] == '1')
20        printf("String is accepted by the CFG.\n");
21    else
22        printf("String is rejected by the CFG.\n");
23
24    return 0;
25 }
```

Output

Enter the string: 1101
String is rejected by the CFG.

=== Code Execution Successful ===

```
main.c
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char str[100];
7     int i, n, valid = 1;
8     printf("Enter the string: ");
9     scanf("%s", str);
10    n = strlen(str);
11    for (i = 0; i < n; i++)
12    {
13        if (str[i] != '0' && str[i] != '1')
14        {
15            valid = 0;
16            break;
17        }
18    }
19    if (valid && n >= 2 && str[0] == '0' && str[n - 1] == '1')
20        printf("String is accepted by the CFG.\n");
21    else
22        printf("String is rejected by the CFG.\n");
23
24    return 0;
25 }
```

Output

Enter the string: 00101
String is accepted by the CFG.

=== Code Execution Successful ===

2.

```
main.c
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char str[100];
7     int i, n, valid = 1;
8
9     printf("Enter the string: ");
10    fgets(str, sizeof(str), stdin);
11
12    str[strlen(str) - 1] = '\0';
13    n = strlen(str);
14
15    for (i = 0; i < n; i++)
16    {
17        if (str[i] != '0' && str[i] != '1')
18        {
19            valid = 0;
20            break;
21        }
22    }
23    for (i = 0; i < n / 2 && valid; i++)
24    {
25        if (str[i] != str[n - 1 - i])
26        {
27            valid = 0;
28            break;
29        }
30    }
31    if (valid)
32        printf("String is accepted by the CFG.\n");
33    else
34        printf("String is rejected by the CFG.\n");
35 }
```

Output

Enter the string: 0110
String is accepted by the CFG.

=== Code Execution Successful ===

```
main.c
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char str[100];
7     int i, n, valid = 1;
8
9     printf("Enter the string: ");
10    fgets(str, sizeof(str), stdin);
11
12    str[strcspn(str, "\n")] = '\0';
13    n = strlen(str);
14
15    for (i = 0; i < n; i++)
16    {
17        if (str[i] != '0' && str[i] != '1')
18        {
19            valid = 0;
20            break;
21        }
22    }
23    for (i = 0; i < n / 2 && valid; i++)
24    {
25        if (str[i] != str[n - i - 1])
26        {
27            valid = 0;
28            break;
29        }
30    }
31    if (valid)
32        printf("String is accepted by the CFG.\n");
33    else
34        printf("String is rejected by the CFG.\n");

```

Output

Enter the string: 0101
String is rejected by the CFG.

=== Code Execution Successful ===

3.

```
main.c
1 #include <stdio.h>
2 #include <string.h>
3 int main()
4 {
5     char str[100];
6     int i, n;
7     int left = 0, right = 0, valid = 1;
8     printf("Enter the string: ");
9     fgets(str, sizeof(str), stdin);
10    str[strcspn(str, "\n")] = '\0';
11    n = strlen(str);
12    while (left < n && str[left] == '0')
13        left++;
14    i = n - 1;
15    while (i >= left && str[i] == '0')
16    {
17        right++;
18        i--;
19    }
20    if (left != right)
21        valid = 0;
22    for (i = left; i < n - right && valid; i++)
23    {
24        if (str[i] != '1')
25        {
26            valid = 0;
27            break;
28        }
29    }
30    if (valid)
31        printf("String is accepted by the CFG.\n");
32    else
33        printf("String is rejected by the CFG.\n");
34    return 0;

```

Output

Enter the string: 0011000
String is rejected by the CFG.

=== Code Execution Successful ===

```
main.c
1 #include <stdio.h>
2 #include <string.h>
3 int main()
4 {
5     char str[100];
6     int i, n;
7     int left = 0, right = 0, valid = 1;
8     printf("Enter the string: ");
9     fgets(str, sizeof(str), stdin);
10    str[strcspn(str, "\n")] = '\0';
11    n = strlen(str);
12    while (left < n && str[left] == '0')
13        left++;
14    i = n - 1;
15    while (i >= left && str[i] == '0')
16    {
17        right++;
18        i--;
19    }
20    if (left != right)
21        valid = 0;
22    for (i = left; i < n - right && valid; i++)
23    {
24        if (str[i] != '1')
25        {
26            valid = 0;
27            break;
28        }
29    }
30    if (valid)
31        printf("String is accepted by the CFG.\n");
32    else
33        printf("String is rejected by the CFG.\n");
34    return 0;

```

Output

Enter the string: 0011100
String is accepted by the CFG.

=== Code Execution Successful ===

4.

Programiz C Online Compiler

Programiz PRO

main.c

↶ ↷ ⚙ ⚡ Share Run

Output

Clear

```
1 #include <stdio.h>
2 #include <string.h>
3 int main()
4 {
5     char str[100];
6     int i, n;
7     int zeros = 0, ones = 0;
8     int valid = 1;
9     printf("Enter the string: ");
10    fgets(str, sizeof(str), stdin);
11    str[strcspn(str, "\n")] = '\0';
12    n = strlen(str);
13    i = 0;
14    while (i < n && str[i] == '0')
15    {
16        zeros++;
17        i++;
18    }
19    while (i < n && str[i] == '1')
20    {
21        ones++;
22        i++;
23    }
24
25    if (i != n || zeros != ones)
26        valid = 0;
27    if (valid)
28        printf("String is accepted by the CFG.\n");
29    else
30        printf("String is rejected by the CFG.\n");
31
32    return 0;
33 }
```

Enter the string: 00111
String is rejected by the CFG.

--- Code Execution Successful ---

main.c

↶ ↷ ⚙ ⚡ Share Run

Output

Clear

```
1 #include <stdio.h>
2 #include <string.h>
3 int main()
4 {
5     char str[100];
6     int i, n;
7     int zeros = 0, ones = 0;
8     int valid = 1;
9     printf("Enter the string: ");
10    fgets(str, sizeof(str), stdin);
11    str[strcspn(str, "\n")] = '\0';
12    n = strlen(str);
13    i = 0;
14    while (i < n && str[i] == '0')
15    {
16        zeros++;
17        i++;
18    }
19    while (i < n && str[i] == '1')
20    {
21        ones++;
22        i++;
23    }
24
25    if (i != n || zeros != ones)
26        valid = 0;
27    if (valid)
28        printf("String is accepted by the CFG.\n");
29    else
30        printf("String is rejected by the CFG.\n");
31
32    return 0;
33 }
```

Enter the string: 000111
String is accepted by the CFG.

--- Code Execution Successful ---

5.

```
main.c
1 #include <stdio.h>
2 #include <string.h>
3 int main()
4 {
5     char str[100];
6     int i, n;
7     int valid = 1;
8     int found = 0;
9     printf("Enter the string: ");
10    scanf("%s", str);
11    n = strlen(str);
12    for (i = 0; i < n; i++)
13    {
14        if (str[i] != '0' && str[i] != '1')
15        {
16            valid = 0;
17            break;
18        }
19    }
20    if (valid)
21    {
22        for (i = 0; i <= n - 3; i++)
23        {
24            if (str[i] == '1' &&
25                str[i + 1] == '0' &&
26                str[i + 2] == '1')
27            {
28                found = 1;
29                break;
30            }
31        }
32    }
33    if (valid && found)
34        printf("String is accepted by the CFG.\n");
35    else
36        printf("String is rejected by the CFG.\n");
37    return 0;
38 }
```

Output

Enter the string: 1101010
String is accepted by the CFG.

=== Code Execution Successful ===

```
Programiz C Online Compiler
main.c
1 #include <stdio.h>
2 #include <string.h>
3 int main()
4 {
5     char str[100];
6     int i, n;
7     int valid = 1;
8     int found = 0;
9     printf("Enter the string: ");
10    scanf("%s", str);
11    n = strlen(str);
12    for (i = 0; i < n; i++)
13    {
14        if (str[i] != '0' && str[i] != '1')
15        {
16            valid = 0;
17            break;
18        }
19    }
20    if (valid)
21    {
22        for (i = 0; i <= n - 3; i++)
23        {
24            if (str[i] == '1' &&
25                str[i + 1] == '0' &&
26                str[i + 2] == '1')
27            {
28                found = 1;
29                break;
30            }
31        }
32    }
33    if (valid && found)
34        printf("String is accepted by the CFG.\n");
35    else
36        printf("String is rejected by the CFG.\n");
37    return 0;
38 }
```

Programiz PRO

Output

Enter the string: 110011
String is rejected by the CFG.

=== Code Execution Successful ===